Why do we need to test a PCBA when there is a functional test at the end of the line or during the box assembly?

ICT (in-circuit testing) has evolved alongside sophisticated technologies, with significant changes in components along the way. A single integrated circuit (IC) functions as a computer, posing challenges in comprehensively testing all its functions. Fortunately, we have developed our testing equipment in adherence to IEEE standards. We have created a test capable of assessing the structural integrity of components from a manufacturing line. In an electronics manufacturing line, every component undergoes testing to ensure proper soldering and correct placement onto the printed circuit board assembly (PCBA). Detecting defects through our in-circuit test (ICT) significantly reduces the cost of product testing. Without ICT, reliance shifts solely to functional testing, which could lead to product failure as functional tests provide limited information; they do not reveal whether the issue lies with a resistor or if there is an open or short circuit.

Complex boards have thousands of test points. How do modern ICT systems manage such complexity?

Consider a motherboard of a server, specifically a high-speed board structured in a 3-by-3 or 2-by-2 configuration. Achieving 100% test coverage for such a board necessitates placing test points, but this is impractical as many sites reside in the middle of the PCB, which are unreachable for testing. Our software introduces a concept termed 'design for test.' The software is utilised to analyse and combine all test features through this process. Leveraging IEEE 1149.1 on the scan, which is a standard in around 80-90% of large application-specific integrated circuits (ASICs) and central processing units (CPUs), allows for comprehensive testing without needing 100% test points on the board.

Essentially, this standard embeds the boundary scan cell like a test point inside the IC, resembling an in-circuit test within the IC itself. The 1149.1 boundary scan software assesses the IC for interconnectivity, eliminating the need for 100% test points on PCBA to achieve high test coverage during ICT testing. With this capability, designers can focus on strategically placing test points at critical locations, reducing the need for unnecessary points. By the time the product reaches mass production, 80-90% of test points have already been covered. This benefits larger boards with up to 10,000 nodes, where the cost of testing can escalate to $1 million per system. Additionally, using fewer test points minimises the risk of false results and mechanical failures, showcasing the evolution of technology to strike a balance in testing processes.

How does that translate into RoI?

The electronics manufacturing test is run with cost sensitivity in mind. By reducing the number of test points, you reduce the cost of the system and almost 50% of the cost of the ICT fixture.

What percentage of test points on a board must be tested—to consider it as a 'tested' board?

Test coverage measures the efficiency, reliability, and quality of an electronics manufacturing test. A test guaranteeing 90% test coverage is a good electronics manufacturing test. However, if it is 50% or less, the board will surely have problems in the end-of-line testing, functional tests, or even when assembled into the product. That's why it's important during prototype or NPI (new product introduction) runs to re-evaluate test coverage to ensure that you have at least 80 to 90%. While you will not be able to achieve 100% test coverage, achieving 80-90% coverage is optimal.

How do ICT systems relate to the optical Inspection machines?

Inspection systems are typically used to identify missing components or ensure proper soldering. In contrast, our system captures defects early in the manufacturing process, allowing for the prompt removal of faulty components. Traditional optical inspection systems may overlook certain defects, especially if the faulty component is positioned at a specific angle or if there are volume-related issues. Additionally, these systems cannot discern whether a component is a 100-ohm or 100-kilo-ohm component.

Can't the latest AOI machines tell if the part is compliant to the specifications?

Technology in inspection is evolving to an extent where it can discern whether a component is smaller but cannot determine whether its value is correct. Suppose a 100-ohm resistance component, which is supposed to be 100 kilo-ohms, is under observation. The AOI will pass the component, and it will become functional. Such products might fail in real-world applications, especially the mission-critical ones. This is why global companies prefer 100% ICT.

What kind of innovations have you brought about in electronics manufacturing tests?

We have many examples to cite. The first 3070 Series 1, Keysight Technologies (HP at that time) introduced the short wire measure-